equation. Both would represent equally sensible ways in which things could evolve.[7]

Of course, the “movie” now referred to is quite different from the ones used in analyzing the motion of a tennis ball or a splattering egg in the last chapter. Probability waves are not things we can see directly; there are no cameras that can capture probability waves on film. Instead, we can describe probability waves using mathematical equations and, in our mind’s eye, we can imagine the simplest of them having shapes such as those in Figures 4.5 and 4.6. But the only access we have to the probability waves themselves is indirect, through the process of measurement.

That is, as outlined in Chapter 4 and seen repeatedly in the experiments above, the standard formulation of quantum mechanics describes the unfolding of phenomena using *two* quite distinct stages. In stage one, the probability wave—or, in the more precise language of the field, the *wavefunction*—of an object such as an electron evolves according to the equation discovered by Schrödinger. This equation ensures that the shape of the wavefunction changes smoothly and gradually, much as a water wave changes shape as it travels from one side of a lake toward the other.[18] In the standard description of the second stage, we make contact with observable reality by measuring the electron’s position, and when we do so, the shape of its wavefunction sharply and abruptly changes. The electron’s wavefunction is unlike more familiar examples like water waves and sound waves: when we measure the electron’s position, its wavefunction spikes or, as illustrated in Figure 4.7, it collapses, dropping to the value 0 everywhere the particle is not found and surging to 100 percent probability at the single location where the particle is found by the measurement.

Stage one—the evolution of wavefunctions according to Schrödinger’s equation—is mathematically rigorous, totally unambiguous, and fully accepted by the physics community. Stage two—the collapse of a wavefunction upon measurement—is, to the contrary, something that during the last eight decades has, at best, kept physicists mildly bemused, and at worst, posed problems, puzzles, and potential paradoxes that have devoured careers. The difficulty, as mentioned at the end of Chapter 4, is that according to Schrödinger’s equation,